

AI MINI – GROUP PROJECT

Artificial Intelligence (AI) and Data Science are transforming industries by enabling smarter, data-driven decision-making. To prepare learners for this rapidly evolving field, this AI Mini-Project Assignment is designed to offer hands-on experience with the end-to-end development of AI solutions.

Through this project, students will work in groups to identify real-world problems, acquire and analyze data, build machine learning models, and interpret results to derive meaningful insights. The mini-project simulates the practical workflow of AI development starting from problem definition and ending with a basic deployment thus encouraging both critical thinking and technical proficiency.

The aim is not only to apply theoretical knowledge but also to cultivate collaborative problem-solving, ethical awareness, and the ability to communicate findings effectively. Each group is expected to document their process and present their work in a clear and professional manner. This exercise is an essential step in bridging the gap between classroom learning and real-world AI applications.

1. Problem Definition & Objective (10 marks)

Guidelines:

- Clearly define the **problem statement**.
- State the **goal** of your project.
- Justify the **relevance** and **potential impact** of solving this problem.
- Specify whether it's a **classification**, **regression**, **clustering**, or other AI task.

Deliverable: One paragraph problem description and project objective.

2. Data Acquisition (10 marks)

Guidelines:

- Source relevant, clean, and useful data.
 - Sources: Kaggle, UCI, government portals, APIs, web scraping, etc.
- Describe the dataset: size, format, number of features, time span, etc.
- Include data collection method if not using a public dataset.

Deliverable: Brief summary of data source, description, and justification.

3. Data Cleaning & Preprocessing (10 marks)

Guidelines:

- Handle missing values, duplicates, and outliers.
- Convert data types where necessary.
- Normalize, encode, or transform variables.
- Split the dataset into training and test sets.

Deliverable: Code + explanation of key cleaning steps.

4. Exploratory Data Analysis (EDA) (10 marks)

Guidelines:

- Use visualizations (histograms, scatterplots, boxplots, heatmaps).

- Identify trends, correlations, and patterns.
- Summarize key statistical findings and feature relationships.

Deliverable: Visuals + insights.

5. Feature Engineering (10 marks)

Guidelines:

- Create or transform variables to improve model performance.
- Encode categorical data appropriately.
- Scale numerical features (if needed).
- Remove irrelevant or redundant features.

Deliverable: Code with explanation of new features or transformations.

6. Model Building (10 marks)

Guidelines:

- Choose one or more suitable models based on the problem type.
 - Examples: Logistic Regression, Decision Trees, SVM, XGBoost, CNN, etc.
- Train models and tune hyperparameters (Grid Search, Random Search).
- Ensure reproducibility (random seeds, clean code).

Deliverable: Trained model + rationale for model choice.

7. Model Evaluation (10 marks)

Guidelines:

- Evaluate using appropriate metrics:
 - Classification: Accuracy, Precision, Recall, F1-Score, ROC-AUC.
 - Regression: MSE, MAE, R-squared.
- Use confusion matrix or evaluation plots.
- Compare performance across models (if applicable).

Deliverable: Performance report and comparison table.

8. Results Interpretation & Insights (10 marks)

Guidelines:

- Identify important features (feature importance, SHAP/LIME if advanced).
- Explain what the results mean in the real-world context.
- Provide actionable insights or recommendations.

Deliverable: Summary of key findings in real-world terms.

9. Deployment (10 marks)

Guidelines:

- Deploy your model using:
 - **Streamlit**, **Flask**, or **Gradio** for a simple web app.
 - Host it locally or on platforms like **Render**, **Hugging Face Spaces**, **Heroku**, or **GitHub Pages**.
- Allow users to upload data or enter values to get predictions.

- Ensure user interface is simple and functional.

Deliverable: Live or local demo + GitHub repo with instructions.

10. Documentation & Presentation (10 marks)

Guidelines:

- Submit a well-documented report including:
 - Introduction, methodology, results, and conclusion.
- Include links to GitHub, deployed model/app, and data sources.
- Present the project (slides) covering:
 - Problem, data, methods, results, demo, and learnings.

Deliverables:

- Final report (PDF or DOCX)
- Presentation slides (PowerPoint or PDF)
- GitHub link

Submission Checklist

- Clear objective and defined problem
- Dataset description and cleaning steps
- EDA with insights
- Feature engineering
- Model code + evaluation
- Insights derived from the model
- Deployment demo or app link
- Documentation and presentation files

- GitHub repo with clean code

SAMPLE TASKS

Here is a List of data science/AI mini-project ideas across various domains and topics:

Predicting Housing Prices: Build a regression model to predict the prices of houses based on features like location, size, number of bedrooms, etc.

Customer Segmentation: Use clustering algorithms to segment customers based on their purchasing behavior, demographics, or other attributes.

Credit Risk Analysis: Develop a classification model to assess the credit risk of loan applicants based on their financial history, demographics, and other relevant factors.

Churn Prediction: Predict customer churn (attrition) for a subscription-based service using historical usage data and demographic information.

Sentiment Analysis: Analyze the sentiment of customer reviews or social media posts using text classification techniques to classify them as positive, negative, or neutral.

Image Classification: Build a deep learning model to classify images into different categories, such as identifying objects or classifying medical images.

Fraud Detection: Develop a fraud detection system using anomaly detection techniques to identify unusual patterns or transactions in financial data.

Healthcare Analytics: Analyze electronic health records (EHR) data to identify patterns, trends, and correlations related to patient outcomes, disease prevalence, or treatment effectiveness.

Movie Recommendation System: Build a recommendation system that suggests movies to users based on their viewing history, ratings, and preferences.

Predictive Maintenance: Develop a predictive maintenance model to forecast equipment failures or maintenance needs based on sensor data and historical maintenance records.

Market Basket Analysis: Analyze transaction data from a retail store to identify frequently co-occurring items and generate product recommendations.

Social Network Analysis: Analyze the structure and dynamics of social networks to identify influential nodes, communities, or patterns of interaction.

Time Series Forecasting: Forecast future values of a time series dataset, such as stock prices, weather data, or demand for a product, using statistical or machine learning models.

Natural Language Generation: Build a model that generates human-like text based on input prompts, such as generating product descriptions or writing news articles.

Anomaly Detection: Detect anomalous behavior or outliers in datasets, such as network traffic, server logs, or manufacturing sensor data, using unsupervised learning techniques.

Customer Lifetime Value Prediction: Predict the future value of customers based on their historical behavior and purchase patterns, helping businesses prioritize marketing efforts.

Image Captioning: Build a model that generates descriptive captions for images, explaining the contents or context depicted in the image.

E-commerce Recommendation System: Develop a recommendation engine for an e-commerce platform, suggesting products to users based on their browsing history, purchase behavior, and preferences.

Predicting Employee Attrition: Build a model to predict the likelihood of employees leaving a company based on factors such as job satisfaction, performance metrics, and tenure.

Climate Change Analysis: Analyze climate data to identify long-term trends, patterns, and correlations related to temperature, precipitation, sea levels, and other environmental factors

NB , YOUR MINI PROJECT IS NOT LIMITED TO THE ABOVE

THIS IS A GROUP PROJECT (MAX FIVE PEOPLE)